

1. INTRODUCTION

Sentiment analysis and emotion recognition are the new basic functions of human-machine interaction systems. Having a machine comprehend not only what a person is uttering, but also the emotion behind them uttering it, has wide real-world applications across chatbots, and customer service support systems. This report gives our reproduction of the paper with the title ‘Multimodal Emotion Recognition and Sentiment Analysis in Multi-Party Conversation Contexts by Farhadipour et al. (2025)’.

The original article suggests a multimodal system, which is a combination of four different data modalities, namely, text, audio, facial expressions, and full-frame video, to categorize emotion into seven classes (anger, disgust, fear, joy, neutral, sadness, surprise) and sentiment into three classes (positive, negative, neutral). The system is tested on the MELD dataset, which is based on the TV series Friends and suggests the conversational problems involving two or more parties in a realistic way.

Our reproduction aims to re-implement every unimodal pipeline to the best of our ability, train each one on a sample of the MELD dataset, and compare them to our own and the benchmarks reported in the original paper. The multimodal fusion experiment is also included. We additionally ran multimodal fusion experiments including two three-modality combinations (Text+Audio+Video and Text+Audio+Face) and the full four modality fusion. The remaining were not replicated due to serious limitations on the amount of memory and computational power available in GPUs. A complete rationale of this choice is presented in the Reproducibility Discussion section.

2. PAPER SUMMARY

Farhadipour et al. (2025) discuss the issue of detection of emotions and sentiment in real-world multi-party conversational data. In contrast to the majority of the previous literature based on the use of the IEMOCAP dataset (recorded in controlled studio circumstances), the authors choose the MELD dataset to capture a more realistic and challenging environment with variable backgrounds, environmental noise, and multiple speakers at the same time.

The system proposed is constructed by integrating four pipelines that process a certain set of data. In the text part, it draws on RoBERTa, which is a transformer-based model that was initially trained on millions of tweets (TweetEval dataset) and subsequently fine-tuned on the MELD dataset to learn more about emotions in dialogues. On audio, the system uses Wav2Vec2, which is a model that learns on raw audio signals by predicting missing parts of speech, and is also fine-tuned on MELD after being trained on multiple emotional speech datasets. In the case of facial expressions, a pipeline named FacialNet is employed. It operates in several stages: TalkNet discovers who is talking, followed by InfoMap clustering to separate faces, then InceptionResNetV1 to discover features in each frame, and lastly the Bidirectional LSTM with attention breaks down these features over time to recognize emotions. Finally, when working with video data, the system employs a custom-built model that features MobileNetV2 to extract visual features and Transformer with local self-attention to comprehend the dynamics of emotions change between frames.

All four modalities are combined into one multimodal vector, which is aggregated on a single Multilayer Perceptron, using feature embeddings. It is reported in the paper that the complete four-modality fusion attains 66.36% accuracy on emotion recognition and 72.15% accuracy on sentiment analysis on the MELD test set, which is an improvement of 2.02% and 2.94% respectively over the most accurate unimodal baseline.

3. IMPLEMENTATION DETAILS

This section presents the implementation setup used while reproduction.

3.1. Text Modality

We applied the text pipeline to RoBERTa-base which was loaded through the Hugging Face Transformers library (AutoModel and AutoTokenizer). We sampled 15% of the MELD dataset, maintaining the original train/dev/test splits. All utterances were tokenized up to 128 tokens, and they were padded and truncated. The final hidden layer CLS token representation was projected by a linear classification head of both emotion (7 classes) and sentiment (3 classes) problems. We ran Cross-Entropy loss and AdamW optimizer with a learning rate of $2e-5$, a linear warmup schedule, which warmed up over 10% of all training steps, a batch size of 16, and 5 training epochs. To be reproducible, a fixed random seed of 42 was used. The most appropriate model was chosen with regards to the accuracy of development sets. After training, we stored and saved CLS token embeddings of all splits to allow multimodal fusion experiments in the future.

3.2. Voice Modality

Facebook/Wav2Vec2-base is used on the audio pipeline through Hugging Face. FFmpeg loads the audio files and decodes them to mono 16kHz PCM float32 waveforms. There is zero padding or truncation of files to a fixed maximum of 8 seconds. The Wav2Vec2Processor takes care of normalization and padding on a batch. The last hidden state is a mean-pooled projection performed using a linear classification head. The training was managed using a learning rate of $1e-4$, a batch size of 8, a linear warmup scheduler, and gradient checkpointing and trained over 5 epochs. Thus, this modal is then fine-tuned on MELD dataset.

3.3. Facial Modality

Our facial expression pipeline uses MTCNN from the facenet-pytorch library for face detection, replacing the TalkNet + InfoMap pipeline described in the original paper, which requires a pre-built character face library not publicly available. MTCNN detects and crops the primary face from each frame to a 160x160 image. If no face is detected in a frame, a zero tensor is substituted. Eight frames are uniformly sampled per video clip. Cropped face frames are passed through InceptionResNetV1 pretrained on VGGFace2 (loaded via facenet-pytorch), which is frozen during training. This produces a 512-dimensional embedding per frame. A Bidirectional LSTM then models the temporal sequence of face embeddings, and an attention pooling layer computes a weighted context vector from the LSTM outputs. This vector is passed through a linear classifier. AdamW with learning rate $1e-4$ and batch size 4 was used for 5 epochs, training only the LSTM, attention, and classifier layers.

3.4. Video Modality

The video pipeline was constructed by using MobileNetV2 (pretrained on ImageNet) as a spatial feature extractor. Each video clip is resampled to 224x224, and has 16 uniformly sampled frames. MobileNetV2 uses a two-layer Transformer encoder (8 attention heads, hidden size 1280) with local self-attention as input to the features of MobileNetV2 through a sinusoidal positional encoding layer. The representation of mean-pooled sequences is grouped by a two-layer MLP ($1280 \Rightarrow 512 \Rightarrow \text{num classes}$). It was trained on 5 epochs with AdamW optimizer and learning rate of $1e-4$ with batch size of 4.

3.5. Multimodal Feature Fusion

Once all four unimodal models are trained, feature embeddings (from the penultimate layer of each model) are saved as PyTorch .pt files indexed by utterance UID. For fusion, embeddings are loaded and matched by UID, and only utterances with features available across all required modalities are used. The matched embeddings are concatenated into a single multimodal vector and fed into a FusionMLP classifier: a three-layer network with hidden sizes 1024 and 512, ReLU activations, and 0.3 dropout. AdamW with a learning rate of $1e-3$ and batch size 32 was used for 5 epochs.

4. EXPERIMENTAL SETUP

This section summarizes the results obtained from the proposed method in the paper.

4.1. Hardware

All the experiments were conducted on a Kaggle Notebook with a standardized setup. Two NVIDIA Tesla T4 GPUs with 15 GB of VRAM (30 GB total) and a 4-core Intel Xeon at 2.00 GHz were used in the system. The system memory available was 30 GB of RAM and the disk memory was up to 57.6 GB. The sessions had a maximum running time of 12 hours. Also, the 19.5 GB output storage limit was a critical constraint that influenced our experimental design. It was implemented with the help of PyTorch and Hugging Face Transformers. The models were all trained and tested during the same session so as to be consistent. The dual GPU system had enough memory capacity to support all the unimodal experiments (up to 20% of splits only) but the video and face modality needed special attention in memory management because of its greater computational and storage requirements.

4.2. Dataset Overview

The MELD (Multimodal EmotionLines Dataset) is a collection of more than 13,000 utterances based on the TV show Friends and annotated with seven emotion categories and three sentiment categories. The initial data has the disadvantage of there is a high degree of class imbalance, where neutral and joy are most common and such feelings as fear and disgust are underrepresented. The computational constraints (15% random subset of the initial MELD data was taken in all experiments) meant that not all experiments could be carried out. This subsample was sampled in each split (train/dev/test) maintaining the original distribution of classes. As a result, our experimental data has the intrinsic class imbalance of MELD, and it is representative of the real-life conditions, but it can affect the per-class performance indicators.

4.2.1. Text Preprocessing

The Utterance column of the CSV annotation files contained text utterances which were extracted. All utterances were converted to lower case, removed whitespaces and missing values were replaced with empty strings. Tokenization was trained on the RoBERTa tokenizer with 128 as the maximum sequence length, truncated longer utterances and padded shorter ones.

4.2.2. Audio Preprocessing

FFmpeg was used to pre-extract all video clips to 16 kHz mono WAV files. In loading data, audio data were loaded using the following parameters: PCM float32 format, 1-channel, 16 kHz sampling rate. All waveforms were padded or truncated to a constant length of 8 seconds (128,000 samples). Normalization and batch-level padding were done by the Wav2Vec2Processor.

4.2.3. Video Preprocessing

We used the raw .mp4 files of the original splits of the MELD. Each video clip was uniformly sampled with OpenCV and each frame converted to RGB from BGR, and resized to 224x224 pixels and normalized with ImageNet statistics (mean=[0.485,0.456,0.406], std=[0.229,0.224,0.225]) to 16 uniformly sampled frames.

4.2.4. Face Preprocessing

For the face modality, we first uniformly sample 8 frames from each video clip using OpenCV. Frames are converted from BGR to RGB color space. Face detection and alignment are performed using MTCNN (Multi-task Cascaded Convolutional Networks) with an output size of 160x160 pixels and a margin of 10 pixels. If no face is detected in a frame, a zero tensor of size 3x160x160 is used as a placeholder. Detected face sequences are cached to disk as .pt files to avoid redundant processing during subsequent training epochs. The extracted face sequences serve as input to the FacialNet model.

4.3. Hyperparameter Configuration

Hyperparameter	Text (RoBERTa)	Audio (Wav2Vec2)	Video (MobileNetV2)	Face (FacialNet)
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning Rate	2e-5	1e-4	1e-4	1e-4
Batch Size	16	8	4	8
Epochs	5	5	5	5
Max Sequence Length	128 tokens	8 sec audio	16 frames	8
LR Scheduler	Linear + Warmup (10%)	Linear + Warmup (10%)	None	None

*Table 1: hyperparameters configuration***4.4. Evaluation Metric**

The primary and the only evaluation metric was classification accuracy (correctly classified samples / total samples) as in the original paper. Separate accuracy is reported in terms of the train, dev and test splits. All evaluations were done using accuracy score and classification report functions of scikit-learn.

5. REPRODUCED RESULTS

This section presents the reproduced results of the study.

5.1. Unimodal Results – Emotion Recognition

Table 2 presents the reproduced unimodal results for the emotion recognition task alongside the original paper's reported test accuracy.

Modality	Model	Train Acc (%)	Dev Acc (%)	Test Acc (%)	Paper Result (%)
Text	RoBERTa-base	81.87	60.12	60.93	64.34
Audio	Wav2Vec2-base	47.19	42.94	48.33	51.49
Video	MobileNetV2 + Transformer	47.22	42.94	48.33	36.14
Face	InceptionResNetV1 + BiLSTM	47.22	42.94	48.33	22.61

*Table 2: Unimodal Emotion Recognition Results (Accuracy %)***5.2. Unimodal Results – Sentiment Analysis**

Table 3 presents the reproduced unimodal results for the sentiment analysis task alongside the original paper's reported test accuracy.

Modality	Model	Train Acc (%)	Dev Acc (%)	Test Acc (%)	Paper Result (%)
Text	RoBERTa-base	81.90	65.85	70.26	69.21
Audio	Wav2Vec2-base	86.61	83.33	85.84	56.20
Video	MobileNetV2 + Transformer	47.16	42.68	48.21	42.51
Face	InceptionResNetV1 + BiLSTM	47.63	43.29	47.69	38.98

*Table 3: Unimodal Sentiment Analysis Results (Accuracy %)***5.3. Multimodal Fusion Results**

Table 4 presents the results of our three multimodal fusion experiments: two three-modality combinations and the full four-modality fusion. Results are shown alongside the closest matching configurations from the original paper.

Config	Modalities	Our Emotion Test (%)	Our Sentiment Test (%)	Paper Emotion (%)	Paper Sentiment (%)
3-Mod	Text + Audio + Video	62.98	88.58	66.29	71.76
3-Mod	Text + Audio + Face	63.24	86.30	66.25	71.84
4-Mod	Text + Audio + Video + Face	62.47	88.13	66.36	72.15

Table 4: Multimodal Fusion Results (Test Accuracy %)

5.4. Key Observations

There are a number of interesting trends observable in these findings. To begin with, the strongest single modality system (i.e., RoBERTa) for emotion detection is text data at 60.93%, which also was found to have the highest accuracy (64.34%) in the paper as a single modality. The approximately 3.4% difference can be explained by using a small sample (15%) as training data and using roberta-base instead of using the TweetEval model specifically designed for this task (as used in the paper).

The second aspect of our project involves performing sentiment analysis with our audio model. Our model has very high-test accuracy of 85.84%, compared to 56.20% reported in the paper. The difference in these results primarily can be attributed to the fact that we used a class balanced training strategy, as opposed to the original paper, which trained with the full dataset (with no class balance) and where the neutral class comprised the overwhelming majority of samples. Therefore, Wav2Vec2's training on a more balanced subset of 15% of the data allowed the model to learn all three classes of sentiment equally well. The video and face unimodal models for emotion recognition both had a 48.33% test accuracy which was well above the paper's results of 36.14% for video and 22.61% for face. There were some differences in the video model due to the use of balanced sample sizes and infrastructure. However, the face unimodal model performed better than was originally written and the best explanation seems to be that MTCNN was used for detecting faces instead of using the FacialNet, which was affected by having different expression labels on the same subject over multiple frames.

The third finding is that our four Modality fusion for emotion is lower than that of the paper by 5.89% (62.47% vs 66.36%). However, both of the three modality combinations outperformed the four-modality fusion in emotions. For example, TAV outperformed it by 0.51% and TAF by 0.69%. This could indicate the facial modalities contribute some noise to the fusion process but not much to the final result when combining them with audio and text where both audio and text worked remarkably well when completing the balance emotion task together.

6. REPRODUCIBILITY DISCUSSION

The existing system uses only the English language, and we can expand this to multilingual emotion recognition, which will make the system more useful and applicable to real-world. Furthermore, the authors used simple feature concatenation method, which could be replaced with cross-modal attention or reliability scoring. This would let the model decide which modality should have more effect on the prediction.

6.1. Differences from Original Results

Our reproduced results are consistent with the original paper's overall findings about text being the most dominant of all unimodal modalities and visual modalities being less dominant. In addition, our results show that multimodal fusion is typically better than the unimodal baseline. In contrast, there are notable differences in each of these areas, which we discuss below.

6.2. Reasons for Discrepancies

The discrepancies between our findings and the results in the original paper can be attributed to a few main factors. For instance, we used a smaller version of the MELD dataset for training. This resulted in a smaller amount of training data for emotion recognition, which may not have reached a maximum capacity and,

therefore, may have decreased performance. At the same time, because the subset still retains the original class distribution (rather than being perfectly balanced), any improvement especially in sentiment accuracy is more likely due to task simplicity and model bias toward majority classes rather than true class balance.

Secondly, there is dissimilarity between pre-training of text. Where the original used a version of RoBERTa that had been further pre-trained on large scale tweet data (TweetEval) and we use of the standard roberta-base model. This type of domain specific text pre-training would similarly account for the observed smaller discrepancies in performance score for text-based emotion.

The third difference is in how we applied additional emotion-based pre-training to the Wav2Vec2 model before fine-tuning: the original model employs additional emotion-related pre-training, while we simply used the basic model to directly do so, resulting in a slight decrease in overall emotion accuracy.

Last but not least, there is also a difference between face and video pipelines. The facial detection model is developed using general face detection and therefore does not have speaker specific tracking and thus will inherently introduce noise into the results. The fact that we have used InceptionResNetV1 in the face model will help to keep the features stable, however. In terms of video, we used full self-attention as opposed to the local attention that is proposed in the paper; this could potentially lead to differences in how temporal patterns are represented.

6.3. Ablation Experiments Not Conducted

The original paper indicates that four types of modality combinations were tested (Voice+Face+Video, Text+Face+Video, Text+Face+Voice, and Text+Video+Voice). We were only able to run two of these combinations (Text+Audio+Video and Text+Audio+Face) due to limitations on our GPU resources. The other combinations (Voice+Face+Video and Text+Video+Face) were not able to be completed because running all of the feature files for many large multi-modal fusion experiments concurrently exceeded the capacity of the GPU resources. This is a known limitation of our ability to reproduce the work, as well as a frequent impediment to user reproduction of computer-based, complex multi-modal systems with limited student hardware resources.

It should be noted that the two multimodal combination tests we performed provide strong evidence for the main finding of adding a text modality is the primary factor in multimodal improvements. The emotion results from the TAV (62.98%) and TAF (63.24%) confirm that the text modality has significantly improved the performances from the Voice+Face+Video combination found in prior literature (48.16%).

6.4. Implementation Challenges

A number of implementation difficulties were realized during the reproduction process. The facial pipeline was one of the most challenging elements because the original version of the FacialNet is based on the TalkNet system, InfoMap clustering, and an existing library of faces of the cast of Friends, which are all closed-source. To overcome this, we substituted the initial method with MTCNN in the general face detection that simplified the pipeline but generated extra noise since it did not filter by the speaker.

Audio preprocessing was another challenge. The original paper has relatively little information besides the usage of Wav2Vec2, and therefore we have introduced an extraction process that is based on FFmpeg and uses fixed-length padding. Although effective, this can cause some slight variation in the input distribution as compared to the original arrangement. Secondly, to have the fusion experiments be properly aligned across modalities, it was necessary to handle the utterance IDs (UIDs) carefully because all the modalities should have the same samples. This involved more work to handle missing files and ensure similar naming of text, audio, video, and face features.

Computationally face processing was very costly. Running MTCNN at every frame was much slower than training, thus we added a caching system to save the detected face tensors in disk after the first pass, without having to recalculate in subsequent epochs. Lastly, small discrepancies in versions of datasets were also problematic. The missing video files in some MELD samples were not filtered and needed to be removed, a fact that can slightly impact reproducibility despite sticking to the same overall approach.

7. CONCLUSION

Our full reproduction of the multimodal emotion recognition and sentiment analysis system proposed by Farhadipour et al. (2025) is presented in this report. We have been able to re-implement and re-train all four unimodal pipelines: RoBERTa on text, Wav2Vec2 on audio, MobileNetV2+Transformer on video, and InceptionResNetV1+BiLSTM on facial expressions, and run multimodal fusion experiments on a 15% subset of the MELD dataset, balanced by classes.

We have validated the core arguments of the paper, the text modality is the most robust unimodal cue, visual modalities are far more difficult to train in multi-party conversational contexts and multimodal fusion is better than single unimodal ones. Our 4-modality fusion on emotion recognition had a test accuracy of 62.47, and the paper had a test accuracy of 66.36, a difference of about 3.9 points explainable by smaller training data, no pre-training using RoBERTa, and a basic face isolation pipeline. In the sentiment analysis, our scores were significantly higher than those of the paper in all settings, mainly because of the class-balanced training approach that favors all three sentiment classes. Another interesting side result of our ablation experiments is that both three-modality combinations (TAV: 62.98%, TAF: 63.24%) showed a slight improvement over fully fusing the four modalities (62.47%) on emotion recognition, indicating that face modality might add slight noise to this particular task and dataset combination.

The accompanying code submission contains all the code implemented, training scripts, and a record of all metrics. This reproduction exercise illustrates the soundness of the original paper results as well as the practical difficulty of reproducing multimodal deep learning systems with complex internal components, under limited computational resources.

8. REFERENCES

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